

```
drawArc( int x, int y, int width, int height, int
startAngle, int arcAngle )
```

You must set the color with a different method.

```
fillArc( int x, int y, int width, int height, int
startAngle, int arcAngle )
```

9.3.7 Drawing Polygons and Polylines

A polygon is a complex, multisided shape. It is a complex, multi-angled line. This draws a closed shape, even if the locations of the points do not make that occur. So, no matter where you place the points, the shape is closed.

```
drawPolygon( int xPoints[], int yPoints[], int points )
drawPolygon also has a constructor that takes another polygon object
drawPolygon( Polygon p )
```

```
fillPolygon( int xPoints[], int yPoints[], int points )
```

For the fill, you have to issue a separate method call. This draws a series of lines that are connected. In this case, if you do not make sure the last point coincides with the first, then the shape enclosed by the polylines is not closed.

```
drawPolyline( int xPoints[], int yPoints[], int points )
```

9.4 Graphics2D

The awt graphics are used for basic html graphics and are not suitable for complex graphics. For example, Class: Graphics lets you draw only a very few specific shapes:

- rectangles,
- ovals and
- polygons.

Class Graphics2D allows you to draw any shape. Also, Graphics2D , in contrast to the awt package, allows you to decide the style of line you wish to draw, including its width and fill properties—known as a line's texture. To start with, all these Geometric shapes are created starting with interface and Shape. Curves and arcs are created using the same interface. To determine the fill style of an object, you use: interface and Paint. To determine the pen style of an object, you use: interface and Stroke.

First of all, notice that class Graphics is the Superclass to Graphics2D. Every time you have used this method:

```
public void paint( Graphics g )
```

you have in fact been using a Graphics2D object.